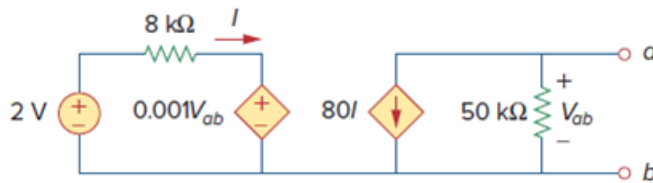


Problem

Obtain the Norton equivalent at terminals a - b of the circuit in Fig. 4.121.

Figure 4.121:



Step-by-step solution

Step 1 of 8

Refer Figure 4.121 in the textbook.

Determine the Norton equivalent resistance across the terminals a and b .

To find R_N , set the independent voltage source impedance equal to zero and connect a current source of $i_o = 1\text{ A}$ to the terminals a and b as shown in Figure 1.

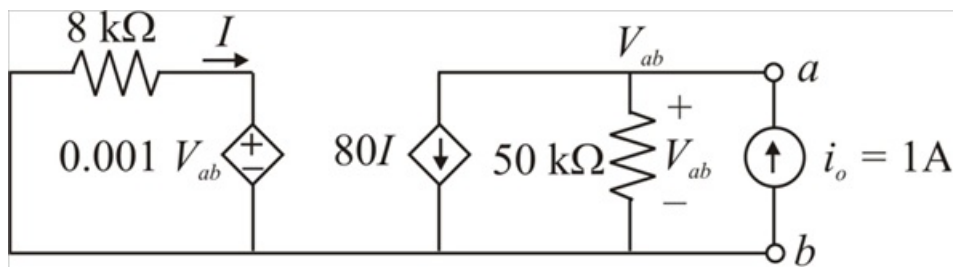


Figure 1

Step 2 of 8

Apply Kirchhoff's Voltage Law to the left side circuit.

$$(8 \times 10^3)I + 0.001V_{ab} = 0$$

$$(8 \times 10^3)I = -0.001V_{ab}$$

$$I = -\frac{0.001V_{ab}}{8 \times 10^3}$$

Thus,

$$I = -(0.125 \times 10^{-6})V_{ab}$$

Step 3 of 8

Apply Kirchhoff's Current Law at node V_{ab} .

$$1 = 80I + \frac{V_{ab}}{50 \times 10^3} \dots\dots (1)$$

Substitute $-(0.125 \times 10^{-6})V_{ab}$ for I in equation (1).

$$1 = 80 \left(-(0.125 \times 10^{-6})V_{ab} \right) + \frac{V_{ab}}{50 \times 10^3}$$

$$1 = \frac{-0.5V_{ab} + V_{ab}}{50 \times 10^3}$$

$$50 \times 10^3 = 0.5V_{ab}$$

Hence,

$$V_{ab} = 100 \times 10^3 \text{ V}.$$

Step 4 of 8

Therefore, the Norton resistance is,

$$\begin{aligned} R_N &= \frac{V_{ab}}{i_o} \\ &= \frac{100 \times 10^3}{1} \\ &= 100 \times 10^3 \end{aligned}$$

Therefore, the value of Norton resistance R_N is 100 kΩ.

Step 5 of 8

To find Norton's current I_N , short-circuit terminals a and b and find the current i_{sc} as shown in Figure 2.

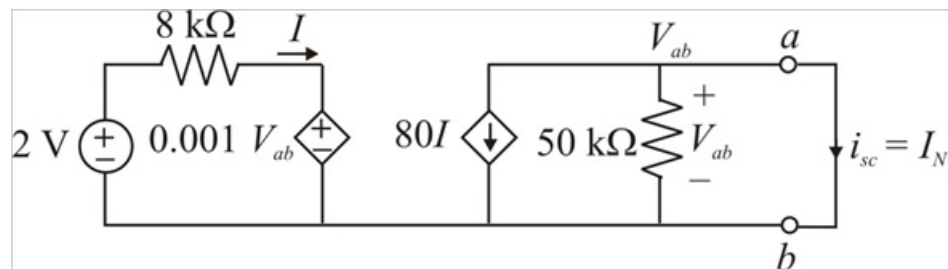


Figure 2

Step 6 of 8

From the circuit shown in Figure 2,

$$V_{ab} = 0V (\because \text{short-circuit})$$

Apply Kirchhoff's Voltage Law to the left side circuit.

$$-2 + (8 \times 10^3)I + 0.001V_{ab} = 0$$

$$-2 + (8 \times 10^3)I + 0.001(0) = 0$$

$$-2 + (8 \times 10^3)I + 0 = 0$$

$$(8 \times 10^3)I = 2$$

$$I = \frac{2}{8 \times 10^3}$$

$$I = 0.25 \times 10^{-3}$$

Therefore, the current, I is 0.25 mA .

Step 7 of 8

Apply Kirchhoff's Current Law at node V_{ab} .

$$80I + \frac{V_{ab}}{50 \times 10^3} + i_{sc} = 0$$

$$80I + \frac{0}{50 \times 10^3} + i_{sc} = 0$$

$$80I + 0 + i_{sc} = 0$$

$$i_{sc} = -80I \dots\dots (2)$$

Substitute 0.25 mA for I in equation (2).

$$i_{sc} = -80(0.25 \times 10^{-3})$$

$$i_{sc} = -20 \times 10^{-3}$$

Therefore, the Norton's current is, $i_{sc} = I_N = -20 \text{ mA}$.

Step 8 of 8

Therefore, the Norton equivalent circuit is shown in Figure 3.

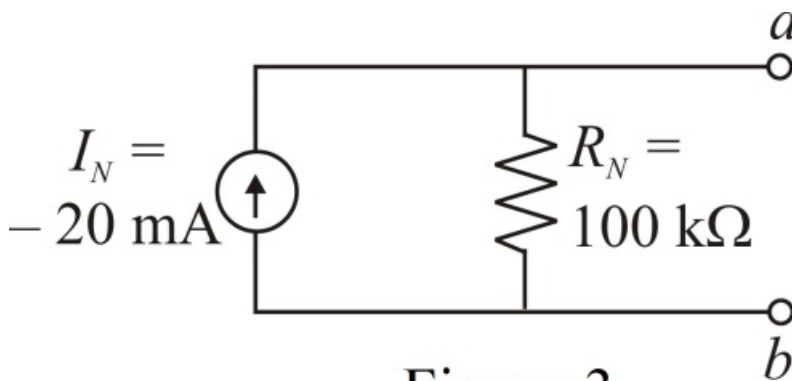


Figure 3